

3-Adapter LoRA Merging Report

Part 1: Three Adapters — Same Dataset, Disjoint Subsets

1.1 Experimental Setup

The central question in this experiment is whether merging three adapters, each trained on one-third of the data, can recover the performance of a single adapter trained on the entire dataset. Three adapters were trained independently on non-overlapping 17K-sample partitions of the Finetome-100k dataset. A fourth model was trained on all 51K samples as the upper-bound reference.

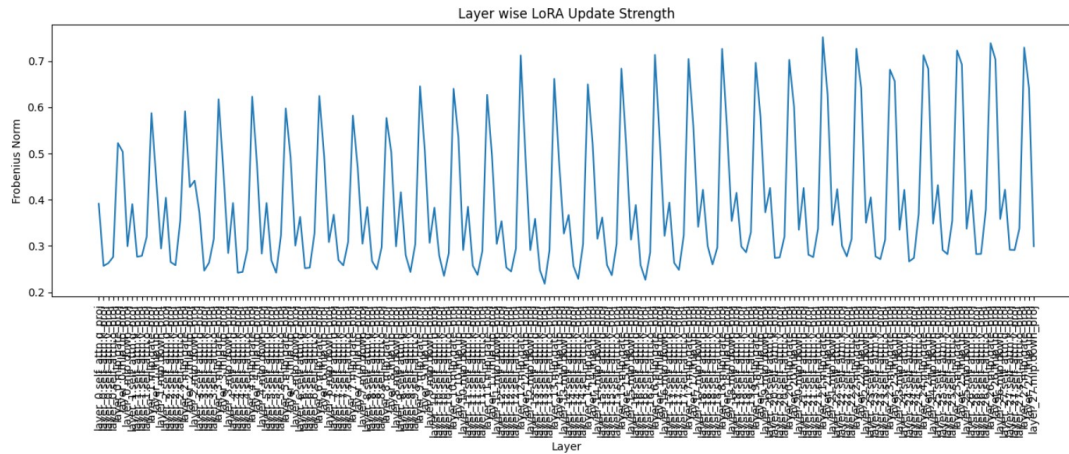
Parameter	Details
Base Model	Qwen3-0.6B
Dataset	Finetome-100k (51K samples used)
Adapter 1	Trained on subset 1 — samples 0 to 17K
Adapter 2	Trained on subset 2 — samples 17K to 34K
Adapter 3	Trained on subset 3 — samples 34K to 51K
Full Model	Trained on all 51K samples (reference baseline)
Merging Goal	Recover full-dataset quality by merging three independently trained subset adapters

1.2 Pre-Merge Analysis

Frobenius Norm — Individual Adapters

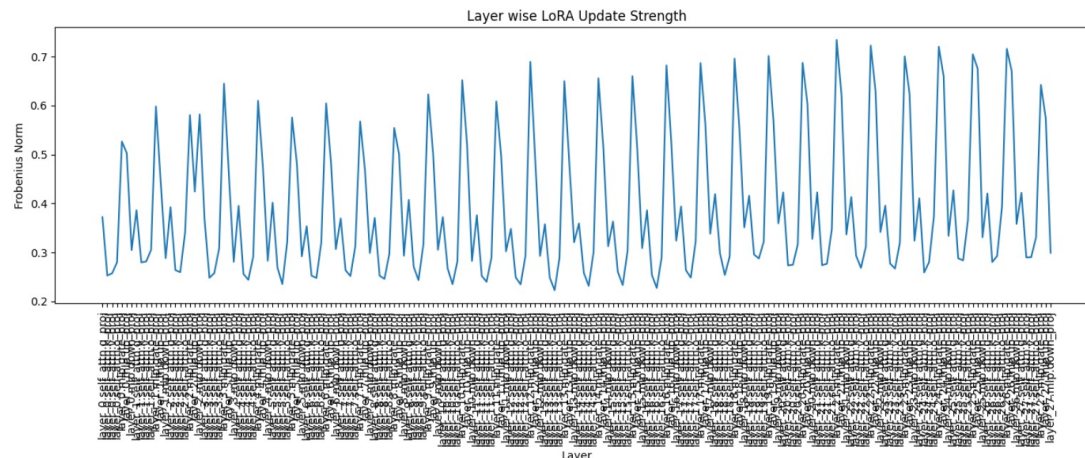
The Frobenius norm of the LoRA delta matrix per layer measures how strongly each adapter modified the base model. Peaks correspond to layers that absorbed the most task-specific information; troughs indicate layers that changed minimally.

Model 1 (Subset 1 — 0 to 17K samples)



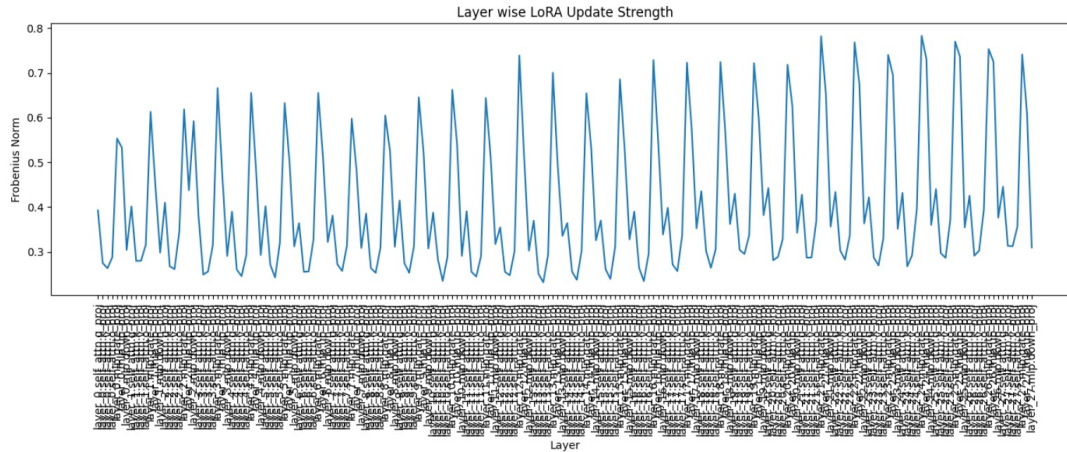
Model 1's update strength oscillates in the 0.2–0.75 range with a clear upward trend toward deeper layers. The pattern is regular and rhythmic, alternating between attention and MLP sub-layers. The growing peak amplitude from block 0 to block 27 indicates that later layers absorbed progressively more task-specific adjustment from the first data partition.

Model 2 (Subset 2 — 17K to 34K samples)



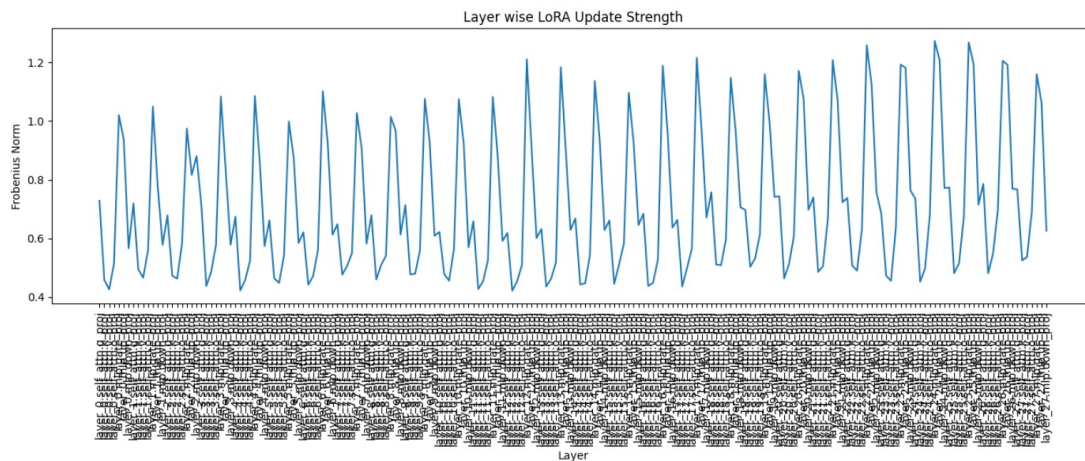
Model 2 shows a very similar profile to Model 1 — norms in the 0.2–0.75 range with the same alternating attention/MLP rhythm and the same upward trend. The close visual match between Models 1 and 2 confirms that different 17K partitions of the same dataset drive nearly identical representational adjustments, which is a strong prerequisite for successful merging.

Model 3 (Subset 3 — 34K to 51K samples)



Model 3 follows the same oscillating pattern with norms between 0.3 and 0.8, slightly higher than Models 1 and 2. The third partition may contain somewhat harder or more diverse examples than the first two, requiring slightly larger adjustments in some layers. The overall structure remains consistent, further supporting the compatibility of the three adapters for merging.

Full Model (All 51K Samples)



The full model shows norms in the 0.4–1.3 range — roughly double the magnitude of the individual subset adapters. This is expected: a model trained on the complete dataset has seen more diverse examples and needs larger total adjustments to encode the full distribution. The gap in norm magnitude between subset adapters (~0.7 peak) and the full model (~1.3 peak) is what the merging methods are trying to bridge.

Effective Rank — All Models



All four models (Models 1, 2, 3, and Full Model) show tightly aligned effective rank profiles across all 28 transformer blocks, generally ranging between 8 and 14 with a shared sharp dip near block_12. The near-perfect overlap of the three subset adapter lines confirms they are exploring nearly identical subspaces per layer. This is the key diagnostic that makes same-dataset merging tractable: if the adapters were using different subspaces, merging would produce destructive interference.

Cosine Similarity Between Subset Adapters and Full Model

The cosine similarity between each subset adapter's weight updates and the full model's updates was measured at 0.62–0.73 across sampled layers. This moderate but meaningful overlap confirms that each 17K partition captures a distinct but complementary portion of the full 51K gradient signal — if similarity were near 1.0, the subsets would be redundant; near 0, merging would be destructive. The 0.62–0.73 range sits in the productive middle ground where merging can genuinely combine information.

1.3 What Is BWSum and Why It Matters

BWSum — Balanced Weighted Subspace Union Merge — is a fundamentally different merging strategy from the others. Rather than combining weight matrices directly (as linear, DARE, and TIES do), it merges adapters in their principal subspace. For each pair of adapters, it decomposes each delta matrix via SVD, allocates a rank budget proportional to the interpolation weight, forms a union of the top subspace directions, orthogonalizes that union via QR decomposition, and reconstructs the merged weight matrix from this unified basis. For three adapters, it applies this process sequentially: merge adapters 1 and 2 first, then merge the result with adapter 3.

The algorithm step by step, for each weight matrix:

1. **SVD each adapter:** $\Delta W_1 \rightarrow U_1, S_1, Vh_1$ and $\Delta W_2 \rightarrow U_2, S_2, Vh_2$
2. **Energy-weighted embedding:** $A_1 = \sqrt{(1-t)} \cdot U_1 \cdot \sqrt{S_1}$, $B_1 = \sqrt{(1-t)} \cdot V_1 \cdot \sqrt{S_1}$
 $A_2 = \sqrt{t} \cdot U_2 \cdot \sqrt{S_2}$, $B_2 = \sqrt{t} \cdot V_2 \cdot \sqrt{S_2}$
3. **Split rank budget:** $r_1 = \text{round}((1-t) \cdot \text{rank})$, $r_2 = \text{rank} - r_1$
4. **Take top directions:** $A_1[:, :r_1]$, $B_1[:, :r_1]$ and $A_2[:, :r_2]$, $B_2[:, :r_2]$
5. **Union:** $A_{\text{union}} = [A_1 | A_2]$, $B_{\text{union}} = [B_1 | B_2]$
6. **QR:** $Q_A, R_A = \text{QR}(A_{\text{union}})$, $Q_B, R_B = \text{QR}(B_{\text{union}})$
7. **Small SVD:** $U_{\sim}, \Sigma, Vh_{\sim} = \text{SVD}(R_A @ R_B^T)$
8. **Reconstruct:** $W_{\text{merge}} = (Q_A @ U_{\sim}) * \Sigma @ (Q_B @ Vh_{\sim})^T$

Why BWSum is Important

- **Operates in subspace rather than weight space.** Unlike Linear, TIES, or DARE, which merge LoRA weight matrices directly, BWSum first decomposes each adapter into its principal subspace using SVD.
- **Captures the most informative directions.** The SVD identifies the dominant singular vectors that encode the most significant task-specific information learned by each adapter.
- **Allocates representation before merging.** Instead of averaging parameters, BWSum assigns each adapter a portion of the merged subspace through a rank budget split.
- **Removes redundant directions.** QR orthogonalization constructs a clean, non-overlapping basis before reconstruction, reducing redundancy in the merged representation.

- **Preserves learned geometry.** The merged adapter is determined by the geometric structure of the learned subspaces rather than the raw magnitudes and signs of individual parameters.

Why BWSum is Balanced

- **Balance is achieved through rank allocation.** An adapter with interpolation weight (t) receives ($r_2 = \text{round}(t \times \text{rank})$) columns, while the remaining adapter receives ($r_1 = \text{rank} - r_2$).
- **Representational capacity is fixed.** The allocated ranks always sum to the original LoRA rank, ensuring that the merged adapter maintains the same total capacity regardless of the interpolation weights.
- **No adapter dominates by default.** An adapter can receive more representational capacity only if it is intentionally assigned a larger interpolation weight.
- **Orthogonalization prevents redundancy.** QR decomposition ensures that the allocated subspace directions are linearly independent, allowing each adapter to contribute unique information within its assigned rank budget.

Why BWSum Underperforms in the Same-Dataset Setting

- **The three adapters learn highly similar representations.** Since each adapter is trained on equally sized partitions of the same dataset, their principal subspaces are nearly identical.
- **The allocated rank budget becomes redundant.** With the default equal weighting ($(t=0.5)$), the rank is split almost equally, but many of the selected singular directions overlap across adapters.
- **QR removes overlapping directions.** During orthogonalization, near-parallel subspace directions collapse into a smaller set of unique basis vectors, reducing the benefit of explicitly forming a subspace union.
- **The resulting merge resembles SVD merging.** Because the source subspaces are already highly aligned, BWSum reconstructs a representation that is very similar to conventional SVD-based merging.
- **This explains its empirical performance.** Consequently, BWSum performs closer to SVD (and DARE in these experiments) than to Linear, which preserves the full averaged weight information without explicitly compressing the shared subspace.

1.4 Merging Methods

Six methods were applied to merge the three subset adapters:

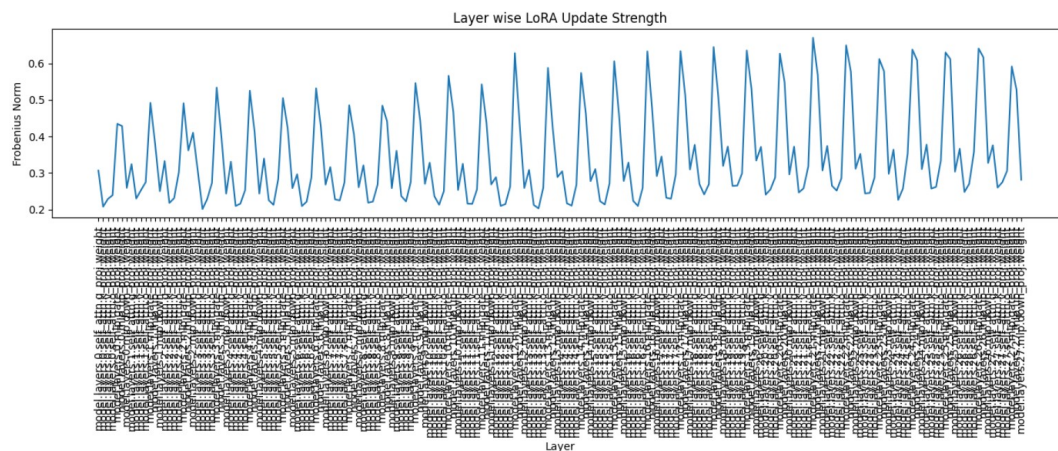
Method	Core Idea	Formula
Linear	Uniform weighted average of all three adapter delta matrices	$\theta^* = \theta_o + (1/3) \cdot \Delta W_1 + (1/3) \cdot \Delta W_2 + (1/3) \cdot \Delta W_3$

SVD	Average deltas then truncate to rank r=16 via Singular Value Decomposition	$\Delta W_{avg} \rightarrow SVD \rightarrow \text{keep top-}r \text{ components} \rightarrow \theta^* = \theta_0 + \Delta W_{merged}$
DARE	Randomly drops 80% of each adapter's delta, rescales survivors by $1/p$, then averages	$\text{mask}(\Delta W_i, p=0.2) \times (1/p) \rightarrow \text{average} \rightarrow \theta^*$
TIES	Trims low-magnitude params, resolves sign conflicts by majority vote, averages only agreed params	$\text{trim} \rightarrow \text{sign vote } \gamma \rightarrow \text{mask } m_i \rightarrow \Sigma(w_i \cdot \gamma_i \cdot m_i) / \Sigma(w_i \cdot m_i) \rightarrow \theta^*$
SLERP	Spherical interpolation along the weight manifold, applied sequentially: $(\Delta W_1, \Delta W_2)$ then with ΔW_3	$SLERP(\Delta W_1, \Delta W_2, 0.5) \rightarrow SLERP(\text{result}, \Delta W_3, 0.5) \rightarrow \theta^*$
BWSum	Balanced Weighted Subspace Union: decomposes each adapter via SVD, allocates rank budget proportional to interpolation weight, concatenates the top subspace directions into a union basis, orthogonalizes via QR, then reconstructs the merged weight from that unified subspace. Applied sequentially for $n > 2$ adapters.	$SVD(\Delta W_i) \rightarrow \text{rank-split budget } (r1, r2) \rightarrow \text{union } [A1 A2], [B1 B2] \rightarrow QR \rightarrow SVD(R_A @ R_B^T) \rightarrow W_{merge} = (Q_A @ U) * \Sigma @ (Q_B @ Vh)^T$

Frobenius Norms of Merged Models

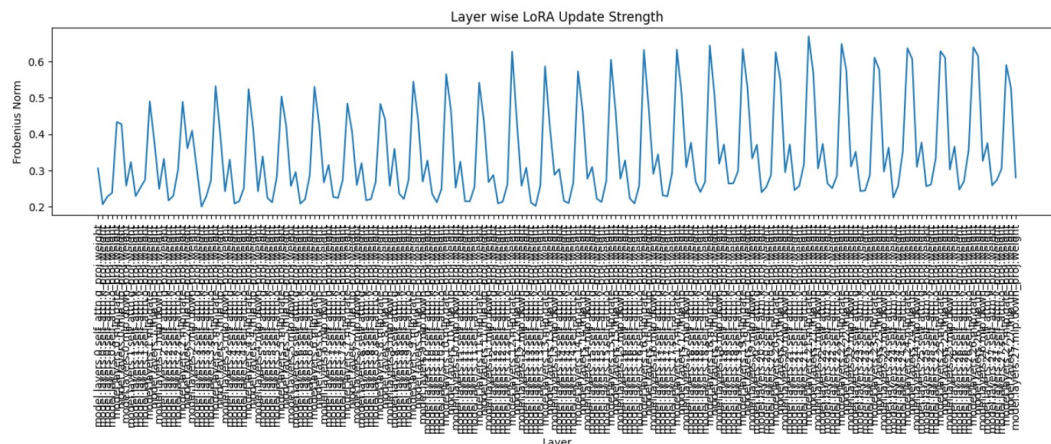
The plots below show the layer-wise LoRA update strength of each merged model. A healthy merged model should show norms comparable to — or ideally approaching — the full model (~0.4–1.3 range).

Linear Merge



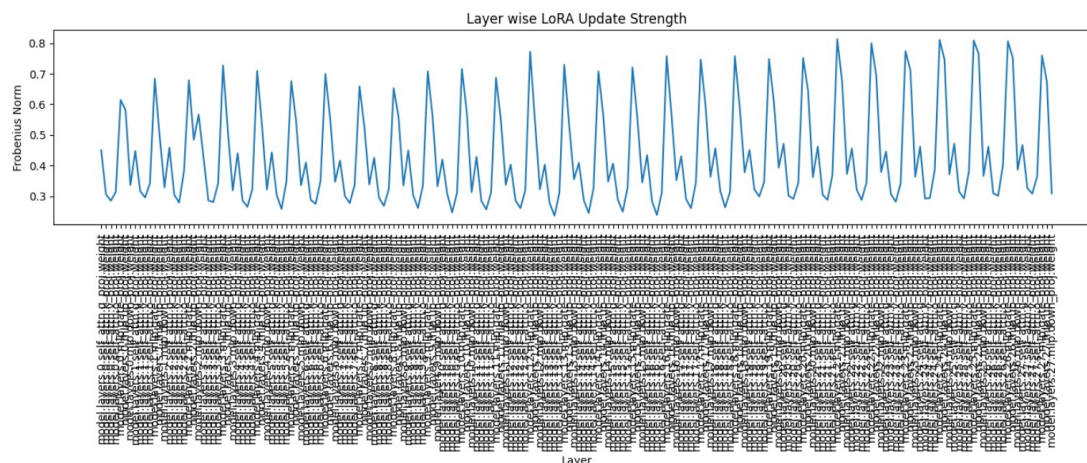
The linear merged model shows norms in the 0.2–0.65 range, following the same alternating attention/MLP rhythm as the individual adapters. The envelope is stable and slightly lower than each individual adapter, consistent with plain averaging attenuating the peaks. While the magnitude is well below the full model, the structural fidelity is high — every layer is represented proportionally.

SVD Merge



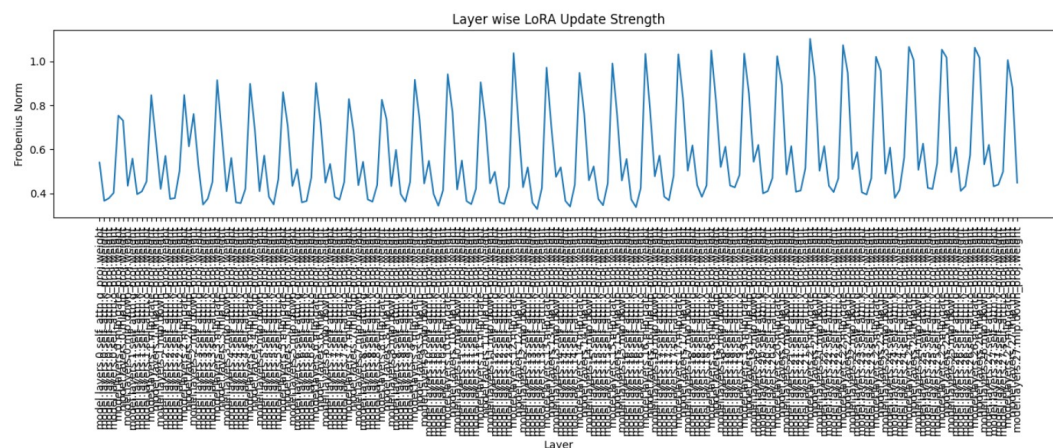
The SVD merged model closely tracks the linear model in norm profile (0.2–0.65 range). The rank-16 truncation removes the smallest singular value components after averaging, but in the same-dataset setting most energy is already concentrated in the top components — so truncation has minimal effect on the overall shape. The result is nearly identical to linear with slightly cleaner oscillation between layers.

TIES Merge



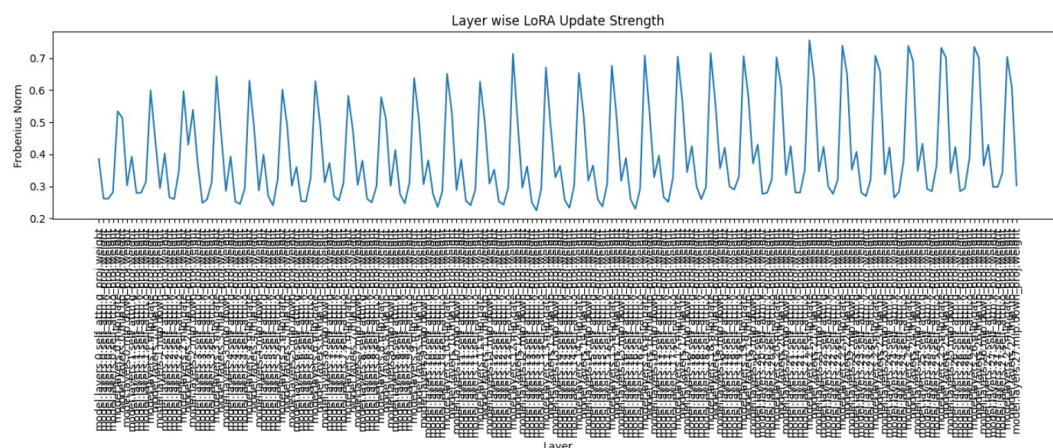
TIES shows norms in the 0.3–0.8 range, somewhat higher than linear and SVD. After trimming low-magnitude parameters and resolving sign conflicts, TIES averages only the parameters where at least two of three adapters agree on direction. In the same-dataset setting, sign agreement is high — almost all parameters across the three adapters point in the same direction — so very few parameters get zeroed out, and the survivors receive a slightly upweighted average, inflating the norm slightly relative to plain linear.

DARE Merge



DARE shows the highest norms of all same-dataset merged models, ranging from 0.4 to 1.1. After randomly dropping 80% of parameters and rescaling survivors by a factor of 5 ($1/p = 1/0.2$), the retained parameters are substantially inflated. In the same-dataset setting where adapter deltas are already modest, this rescaling overcorrects and produces norms well above what the signal warrants — explaining DARE's worst-in-class accuracy.

SLERP Merge



SLERP shows norms in the 0.3–0.75 range with the characteristic upward trend toward deeper layers that geodesic interpolation tends to produce. Applied sequentially — first merging adapters 1 and 2, then merging the result with adapter 3 — SLERP naturally biases toward the angular midpoint of all three adapter directions. The result is structurally sound and achieves the best ROUGE-L in this experiment.

1.5 Evaluation Results

All merged models were evaluated on the Finetome test set using ROUGE-1 (R-1), ROUGE-2 (R-2), and ROUGE-L (R-L). The full model trained on all 51K samples serves as the upper-bound reference.

Model	ROUGE-1	ROUGE-2	ROUGE-L
Qwen3-0.6B-linear-3-adapters-same-dataset-	0.3416	0.0832	0.1673

merged			
Qwen3-0.6B-slerp-3-adapters-same-dataset-merged	0.3370	0.0846	0.1676
Qwen3-0.6B-ties-3-adapters-same-dataset-merged	0.3361	0.0836	0.1673
Qwen3-0.6B-svd-3-adapters-same-dataset-merged	0.3368	0.0820	0.1667
Qwen3-0.6B-dare-3-adapters-same-dataset-merged	0.3278	0.0778	0.1626
Qwen3-0.6B-bwsum-3-adapters-same-dataset-merged	0.3301	0.0782	0.1630
full_finetome (reference)	0.3389	0.0872	0.1703

1.6 Key Observations

- Bootstrap t-test result — no statistically significant differences between any methods: a pairwise bootstrap t-test with 10,000 resamples across all ROUGE metrics found no significant differences between any pair of merged models (all p-values > 0.05). This is the most important finding of Part 1. The ROUGE score differences — Linear 0.3416, SLERP 0.3370, TIES 0.3361, SVD 0.3368, DARE 0.3278, BWSum 0.3301 — are all within random evaluation noise. The correct interpretation is not that Linear is better than DARE, but that all merging methods perform equivalently when adapters are trained on the same task. Method choice does not matter in the same-dataset setting.
 - The primary conclusion is that merging works in general: all six merged models perform in the same range as the full model (ROUGE-1: 0.3389). Three adapters trained on 17K samples each, when merged by any method, approximately recover the performance of a single adapter trained on all 51K samples. This validates data-parallel adapter merging as a practical training strategy regardless of which merge method is chosen.
 - Linear appears numerically best (ROUGE-1: 0.3416) but this is not statistically meaningful. The appearance of marginal superiority is consistent with ensemble averaging reducing SGD optimization noise — but the bootstrap confirms this difference cannot be distinguished from chance under the current evaluation size.
 - DARE shows the lowest scores (ROUGE-1: 0.3278) and is the most likely candidate for genuine underperformance, since its random parameter dropping discards informative signal in the same-dataset setting where all three adapters agree on parameter directions. However, even this gap is not statistically significant at $p < 0.05$.
 - BWSum also underperforms Linear numerically (ROUGE-1: 0.3301). With three equally-sized 17K partitions, all adapters have nearly identical subspace directions, making BWSum's rank budget split allocate columns to near-redundant directions. The result is close to SVD rather than Linear — but again, not significantly different from any other method.
 - The full model retains a numerical advantage on ROUGE-2 (0.0872 vs 0.0832 for the best merge). Bigram overlap is sensitive to sequential co-occurrence statistics built up through continuous 51K-sample training, and this gap likely reflects a genuine structural limitation of data-parallel merging rather than noise — though a larger test set would be needed to confirm significance.
-

Part 2: Three Adapters — Different Tasks, Different Datasets

2.1 Experimental Setup

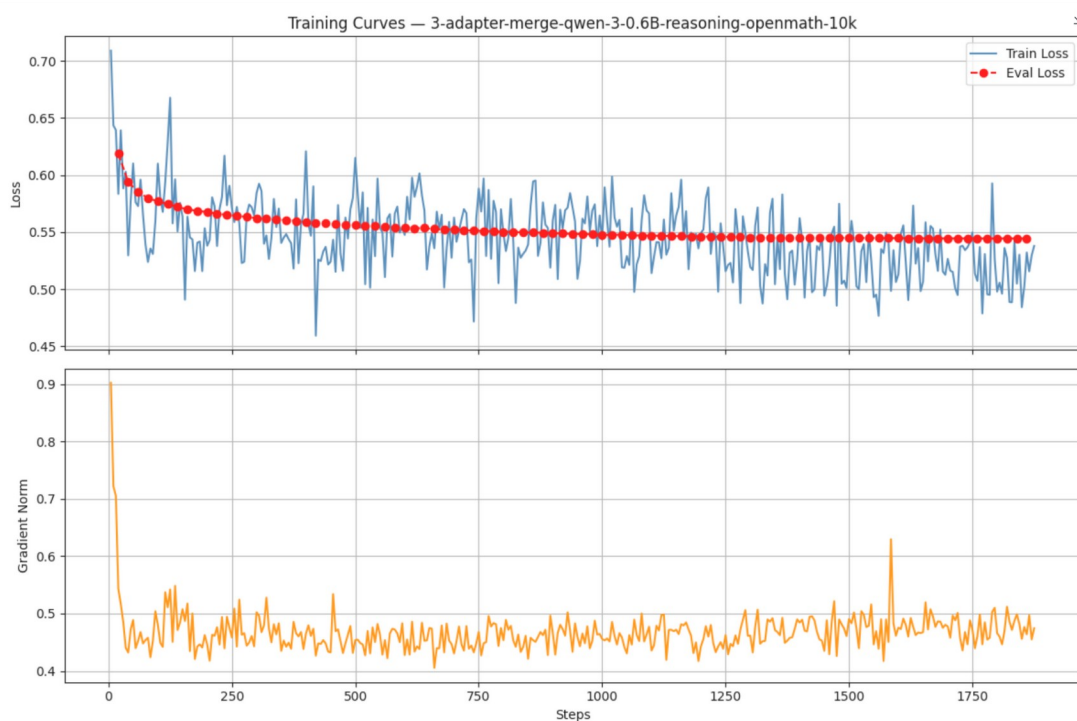
This experiment is substantially more challenging. Three adapters are trained on semantically unrelated tasks: news classification, mathematical reasoning, and instruction following. The goal is a single merged model that retains all three capabilities simultaneously.

Parameter	Details
Base Model	Qwen3-0.6B
Adapter 1 — AG News	Trained on AG News: 4-class news topic classification (World, Sports, Business, Sci/Tech)
Adapter 2 — OpenMath	Trained on OpenMathReasoning: mathematical chain-of-thought reasoning and step-by-step problem solving
Adapter 3 — FineTome	Trained on FineTome: general instruction-following and conversational responses
Merging Goal	Produce a single model that retains all three capabilities simultaneously

2.2 Pre-Merge Analysis

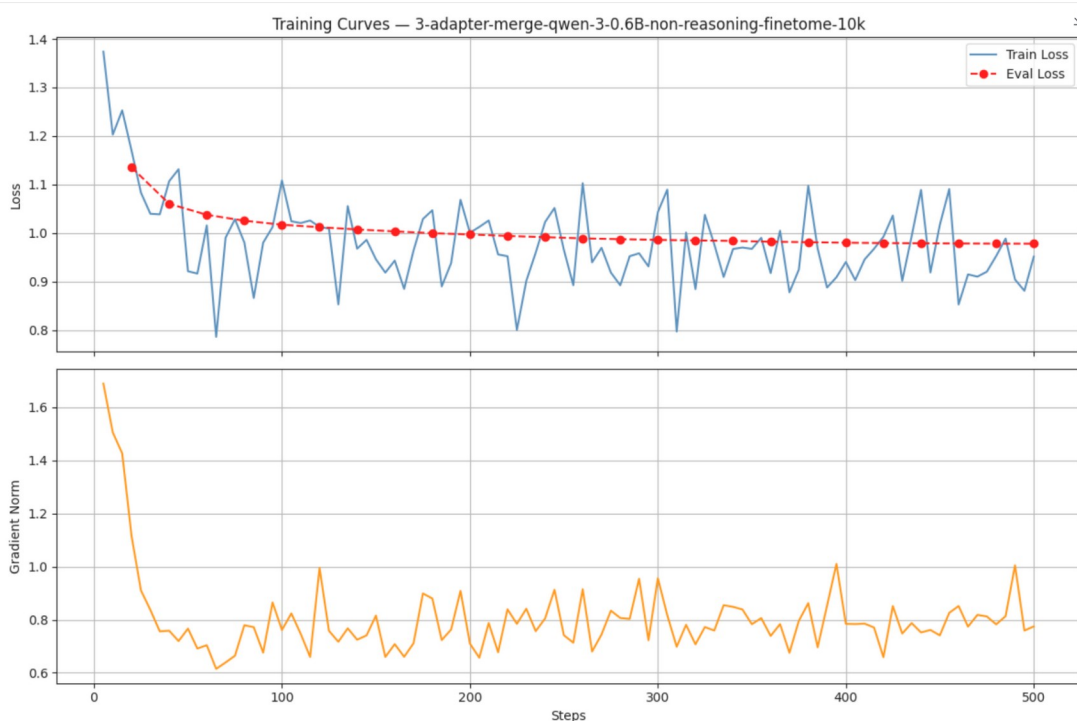
Frobenius Norms — Individual Adapters

OpenMath Adapter



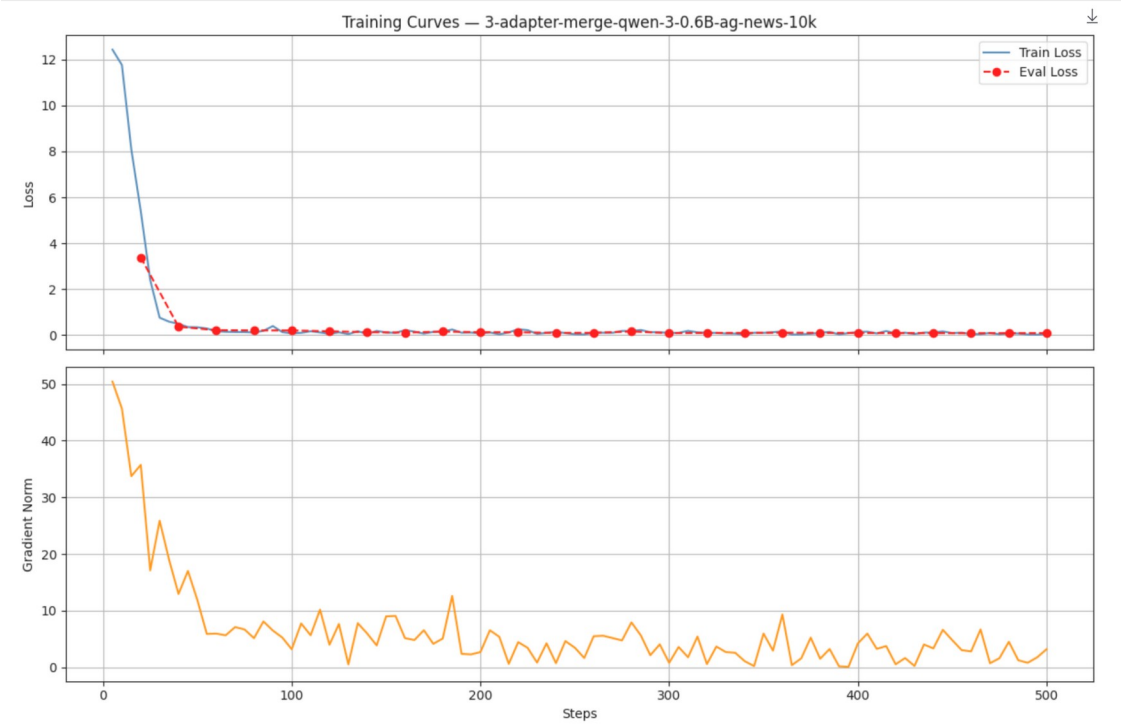
The OpenMath adapter shows update strength ranging from about 0.2 to 0.8, with a gradual upward trend across network depth and the largest peaks (~ 0.6 – 0.78) in the later layers. This reflects the nature of mathematical reasoning: abstract numerical and symbolic computation requires large, concentrated changes in the model's deeper representational layers, while earlier layers handling general syntactic structure are left relatively unchanged.

FineTome Adapter



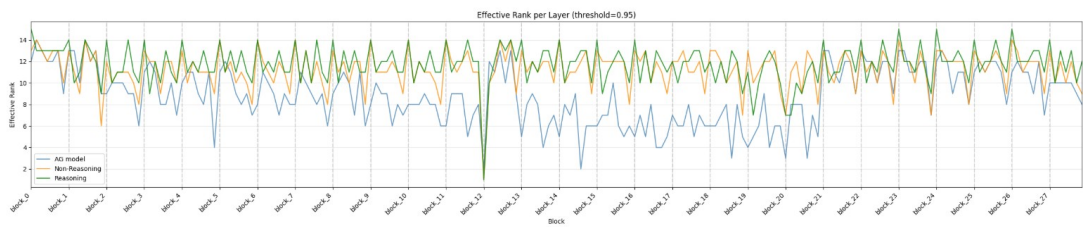
The FineTome adapter shows a much smaller overall range — roughly 0.13 to 0.35 — with a mild upward trend. The consistently lower magnitude compared to OpenMath reflects the nature of general instruction-following: it requires lighter, more distributed adjustments across the entire network rather than large concentrated changes. The model needs to become a better conversationalist, not develop fundamentally new reasoning circuits.

AG News Adapter



Most layers in the AG News adapter show small, flat updates in the 0.1–0.3 range, with one pronounced spike near the very end of the network reaching about 0.7. This is the signature pattern of a classification task: the majority of the network’s representations are left largely unchanged, while a small number of late layers closer to the classification head absorb a disproportionately large share of the adjustment needed to produce discrete class labels.

Effective Rank — All Three Adapters



Unlike the same-dataset case, the three task-specific adapters show clearly differentiated effective rank profiles. The AG News adapter (blue) sits consistently lower — between 5 and 14 — with a pronounced collapse near block_12 and several secondary dips. The OpenMath (green) and FineTome (orange) adapters track each other more closely, generally remaining in the 10–14 range. The divergence of the AG News rank profile is a direct consequence of its classification nature: it uses low-dimensional updates concentrated in specific layers, while the reasoning and instruction-following adapters require richer, higher-rank representations throughout the network.

Cosine Similarity Between Adapters

Because the three adapters were trained on completely different tasks, their weight update directions are largely independent. The cosine similarity analysis quantifies this.

OpenMath vs FineTome

These two adapters show positive but modest cosine similarity, ranging from ~0.04 to ~0.25 across sampled layers:

Layer	Cosine Similarity (OpenMath vs FineTome)
layer_0.mlp.down_proj	0.0533
layer_0.mlp.gate_proj	0.0408
layer_0.mlp.up_proj	0.0780
layer_0.self_attn.k_proj	0.0696
layer_0.self_attn.o_proj	0.0734
layer_0.self_attn.q_proj	0.0762
layer_0.self_attn.v_proj	0.1441
layer_1.self_attn.v_proj	0.2251
layer_10.self_attn.v_proj	0.2549
layer_10.self_attn.q_proj	0.1421
layer_10.self_attn.k_proj	0.1323

The positive values confirm the two adapters point in broadly similar directions — both tasks require the model to understand and generate structured text — but the small magnitudes mean they are not collinear. The v_proj layers show the highest alignment, suggesting that value projections in self-attention adapt similarly for both reasoning and instruction-following tasks.

AG News vs OpenMath and FineTome

The AG News adapter's cosine similarity with the other two adapters is close to zero and frequently negative, indicating near-orthogonality:

Layer	Cosine Similarity (AG News vs Others)
layer_0.mlp.down_proj	0.0019
layer_0.mlp.gate_proj	0.0085
layer_0.mlp.up_proj	-0.0018
layer_0.self_attn.k_proj	0.0026

layer_0.self_attn.o_proj	0.0015
layer_0.self_attn.q_proj	0.0220
layer_0.self_attn.v_proj	-0.0038
layer_1.self_attn.v_proj	0.0350
layer_10.mlp.down_proj	-0.0074
layer_10.mlp.gate_proj	-0.0014
layer_10.mlp.up_proj	0.0007

Values an order of magnitude lower than the OpenMath/FineTome comparison, with sign flips between layers. AG News learned an update direction that is essentially independent of both reasoning and instruction-following. This orthogonality makes cross-task merging more difficult: each method must somehow combine three adapter directions, two of which are modestly aligned and one of which is nearly perpendicular to the others.

2.3 Merging Methods

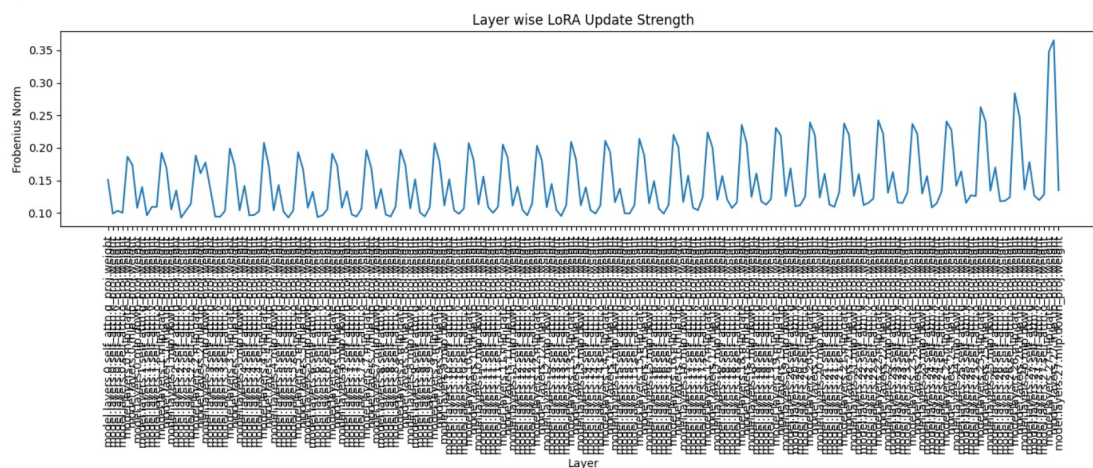
The same six methods applied in Part 1 were used here. In the cross-task setting, the challenges are fundamentally different: adapter deltas are larger, more diverse, and partially conflicting, requiring each method to handle genuine directional disagreement between adapters.

Method	Core Idea	Formula
Linear	Uniform weighted average of all three adapter delta matrices	$\theta^* = \theta_0 + (1/3) \cdot \Delta W_1 + (1/3) \cdot \Delta W_2 + (1/3) \cdot \Delta W_3$
SVD	Average deltas then truncate to rank r=16 via Singular Value Decomposition	$\Delta W_{avg} \rightarrow SVD \rightarrow \text{keep top-}r \text{ components} \rightarrow \theta^* = \theta_0 + \Delta W_{merged}$
DARE	Randomly drops 80% of each adapter's delta, rescales survivors by $1/p$, then averages	$mask(\Delta W_i, p=0.2) \times (1/p) \rightarrow \text{average} \rightarrow \theta^*$
TIES	Trims low-magnitude params, resolves sign conflicts by majority vote, averages only agreed params	$trim \rightarrow \text{sign vote } \gamma \rightarrow \text{mask } m_i \rightarrow \Sigma(w_i \cdot \tau_i \cdot m_i) / \Sigma(w_i \cdot m_i) \rightarrow \theta^*$
SLERP	Spherical interpolation along the weight manifold, applied sequentially: $(\Delta W_1, \Delta W_2)$ then with ΔW_3	$SLERP(\Delta W_1, \Delta W_2, 0.5) \rightarrow SLERP(result, \Delta W_3, 0.5) \rightarrow \theta^*$
BWSum	Balanced Weighted Subspace Union: decomposes each adapter via SVD, allocates rank budget proportional to interpolation weight, concatenates the top subspace directions into a union basis, orthogonalizes via QR, then reconstructs the merged weight from that unified subspace. Applied sequentially for $n > 2$ adapters.	$SVD(\Delta W_i) \rightarrow \text{rank-split budget } (r1, r2) \rightarrow \text{union } [A1 A2], [B1 B2] \rightarrow QR \rightarrow SVD(R_A @ R_B^T) \rightarrow W_{merge} = (Q_A @ U) * \Sigma @ (Q_B @ Vh)^T$

Frobenius Norms of Merged Models — Different Tasks

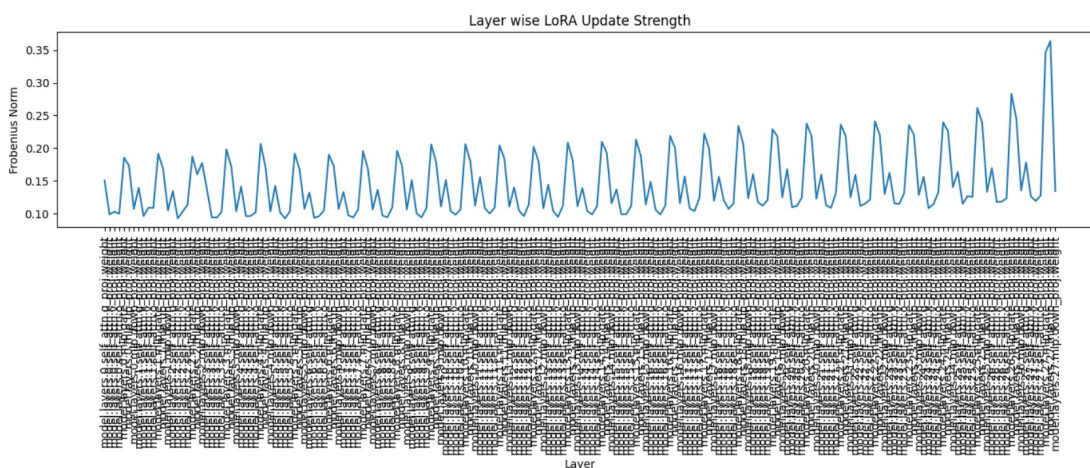
In the cross-task setting, merged model norms are substantially higher than in Part 1, reflecting the larger and more diverse source adapter deltas.

Linear Merge



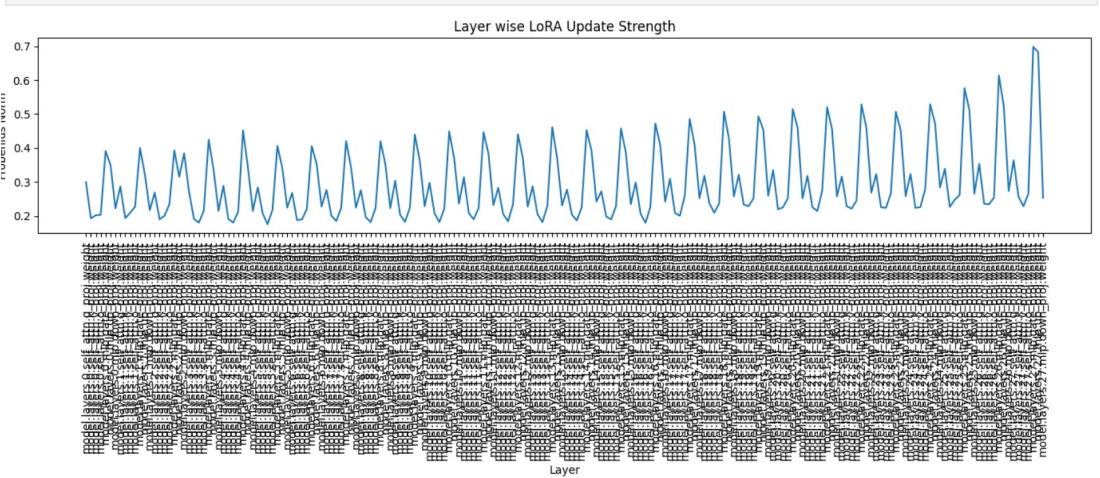
The linear merge of three cross-task adapters produces norms in the 0.10–0.37 range, with one sharp spike near the final layer reaching 0.37 — the AG News classification spike propagating into the merged model. The overall norms are low because linear averaging of three largely orthogonal vectors produces a result significantly smaller in magnitude than any individual adapter. The two reasoning/instruction adapters partially cancel each other, and the AG News adapter contributes mainly at the final spike.

SVD Merge



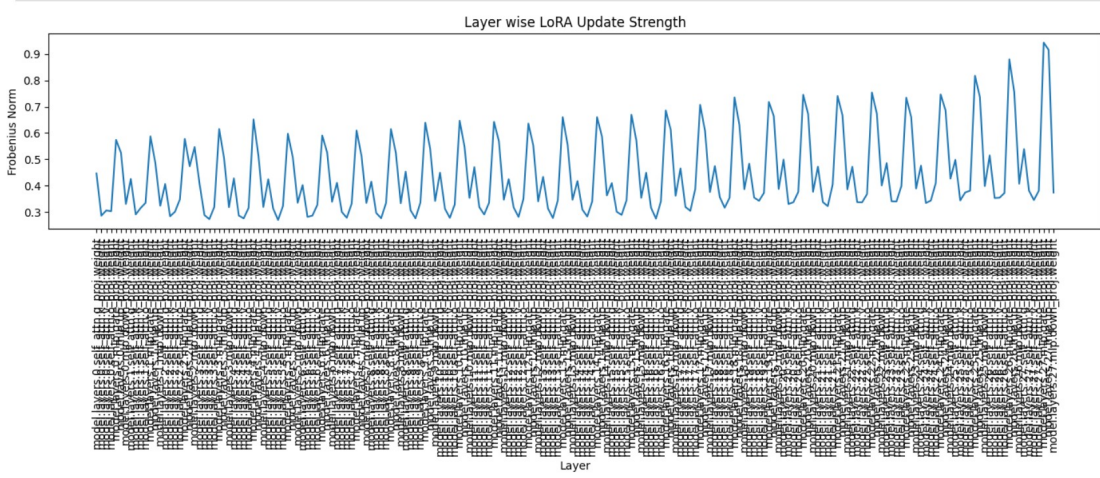
The SVD merged model shows a nearly identical profile to linear (0.10–0.37 range, same final spike). In the cross-task setting, low-rank truncation has similarly limited impact: the merged delta already concentrates its energy in few components because orthogonal source adapters produce a low-rank combined direction by construction. SVD merely confirms that the top-16 singular components account for essentially all the meaningful signal.

TIES Merge



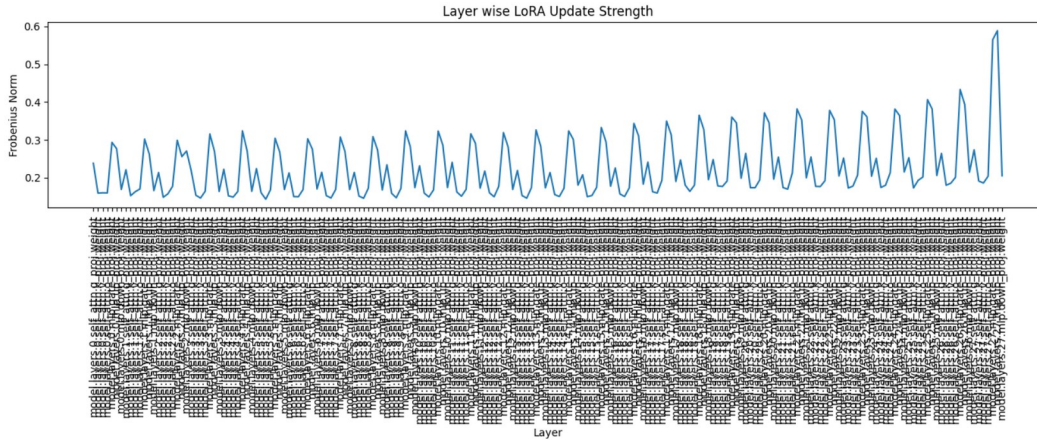
TIES shows higher norms than linear and SVD, ranging from 0.2 to 0.7 with a pronounced upward trend and the sharpest final spike of all methods. The cross-task setting generates extensive sign conflicts between adapters, particularly where AG News conflicts with the other two. TIES zeros out conflicting parameters and then rescales the survivors — the combination of selective retention and rescaling produces higher norms in the layers where at least two adapters agree, and the final spike is amplified because all three adapters agree on the direction of the classification-critical final layers.

DARE Merge



DARE shows norms in the 0.30–0.95 range, growing steadily toward later layers. Unlike the same-dataset case where DARE's rescaling was destructive, here the cross-task adapter deltas are larger and more varied, providing more genuine signal for the 20% of survivors to carry after rescaling. The growing norms toward deeper layers indicate that DARE is preserving more of the task-specific information in the later layers where it is most concentrated.

SLERP Merge



SLERP shows norms in the 0.17–0.60 range, with a steady upward growth toward later layers and the characteristic sharpest spike at the final blocks. The geodesic interpolation maintains the angular relationships between the three adapters — applying sequentially (adapters 1+2 first, then merging with adapter 3) — which means the final merged direction is a careful geometric average of all three adapters' unit-normalised directions, scaled by interpolated magnitudes. This structural care explains SLERP's superior preservation of the AG News update direction.

2.4 Evaluation Results

Evaluated on OpenMath and FineTome Tasks

Each merged model was evaluated on both the mathematical reasoning benchmark (Exact Match) and a text generation task (ROUGE scores). This dual evaluation tests whether the merge preserves reasoning precision and generation fluency simultaneously.

Model	Exact	R-1	R-2	R-L
linear-3-adapters-merged	0.1200	0.3088	0.0623	0.1465
svd-3-adapters-merged	0.1400	0.3090	0.0626	0.1488
ties-3-adapters-merged	0.0900	0.3294	0.0739	0.1595
dare-3-adapters-merged	0.1050	0.3264	0.0721	0.1578
slerp-3-adapters-merged	0.0900	0.3160	0.0676	0.1529
bwsum-3-adapters-merged	0.0950	0.3105	0.0649	0.1489
FineTome adapter (ref)	0.0750	0.3360	0.0816	0.1671
OpenMath adapter (ref)	0.1350	0.3127	0.0688	0.1517

Evaluated on AG News Classification

Each merged model was additionally evaluated as a 4-class news topic classifier. This directly tests whether the AG News classification capability survives the cross-task merge.

Model	Accuracy	Macro F1
slerp-3-adapters-merged	0.5700	0.5676
linear-3-adapters-merged	0.5660	0.5612
bwsum-3-adapters-merged	0.5660	0.5530
dare-3-adapters-merged	0.5480	0.5415
svd-3-adapters-merged	0.5620	0.5597
ties-3-adapters-merged	0.5140	0.5039
Full model (ref — AG News only)	0.5300	0.5412

2.5 Cosine Similarity: Merged Deltas vs AG News

To understand why SLERP outperforms all other methods on AG News, the cosine similarity between each merged model's final weight update and the original AG News adapter's update was measured layer by layer:

Layer	Linear	SLERP	DARE	TIES	SVD
layers.0.mlp.down_proj	0.3557	0.5332	0.1628	0.2705	0.3473
layers.0.mlp.up_proj	0.3814	0.5172	0.1769	0.2942	0.3785
layers.0.self_attn.k_proj	0.4695	0.5340	0.2171	0.3751	0.4674
layers.0.self_attn.q_proj	0.4274	0.5252	0.1960	0.3396	0.4235
layers.0.self_attn.v_proj	0.3693	0.5272	0.1742	0.2883	0.3664
layers.1.mlp.up_proj	0.4323	0.5332	0.2053	0.3363	0.4260
layers.1.self_attn.v_proj	0.3718	0.5387	0.1826	0.2844	0.3662
layers.10.self_attn.q_proj	0.3836	0.5358	0.1807	0.3052	0.3788
layers.10.self_attn.v_proj	0.4091	0.5246	0.1985	0.3327	0.4061
layers.11.mlp.down_proj	0.3031	0.5109	0.1394	0.2374	0.2977

SLERP consistently achieves the highest cosine similarity to the original AG News adapter at every sampled layer (~0.51–0.54), while DARE shows the lowest (~0.14–0.22). Linear and SVD cluster in the middle (~0.30–0.47), with TIES slightly above. This directly explains the AG News accuracy ranking: the merge method that best preserves the direction of the original AG News update achieves the best classification performance, and the one that diverges from it most performs worst.

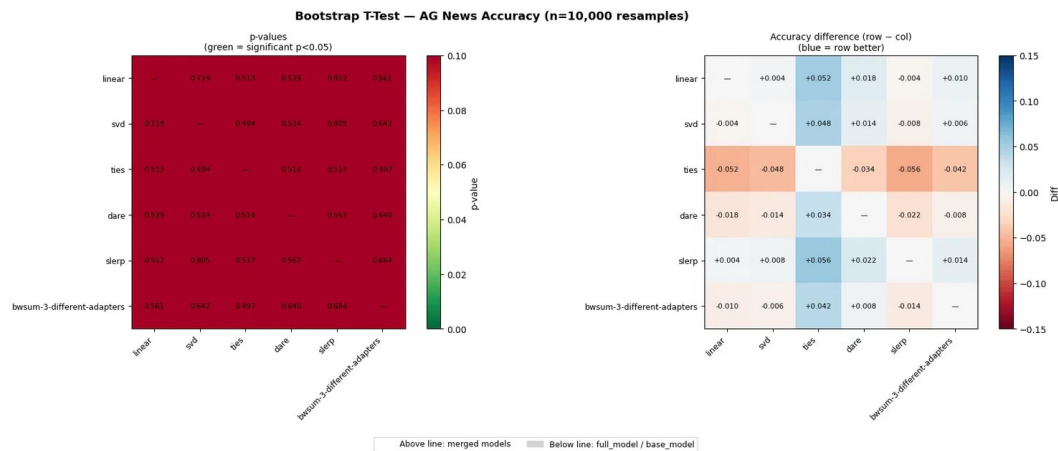
2.6 Key Observations and Statistical Significance

Bootstrap T-Test — AG News Classification

Pairwise bootstrap t-test across all 6 merged models, 10,000 resamples, per-sample accuracy:

- All p-values between 0.49 and 0.91 — no pair is statistically significant.

- No merged model is provably better than any other at classifying news topics at this test set size.
- TIES is the consistent direction-loser: -0.052 vs linear, -0.048 vs svd, -0.056 vs slerp, -0.042 vs bwsum. Every pairwise comparison points negative. This is a strong directional signal even without significance.
- SLERP is the consistent direction-winner: positive vs every other method except bwsum (-0.014). $+0.056$ vs TIES is the largest gap in the experiment.
- Linear, SVD, and BWSum cluster tightly together (differences of ± 0.004 to ± 0.010) — effectively indistinguishable.

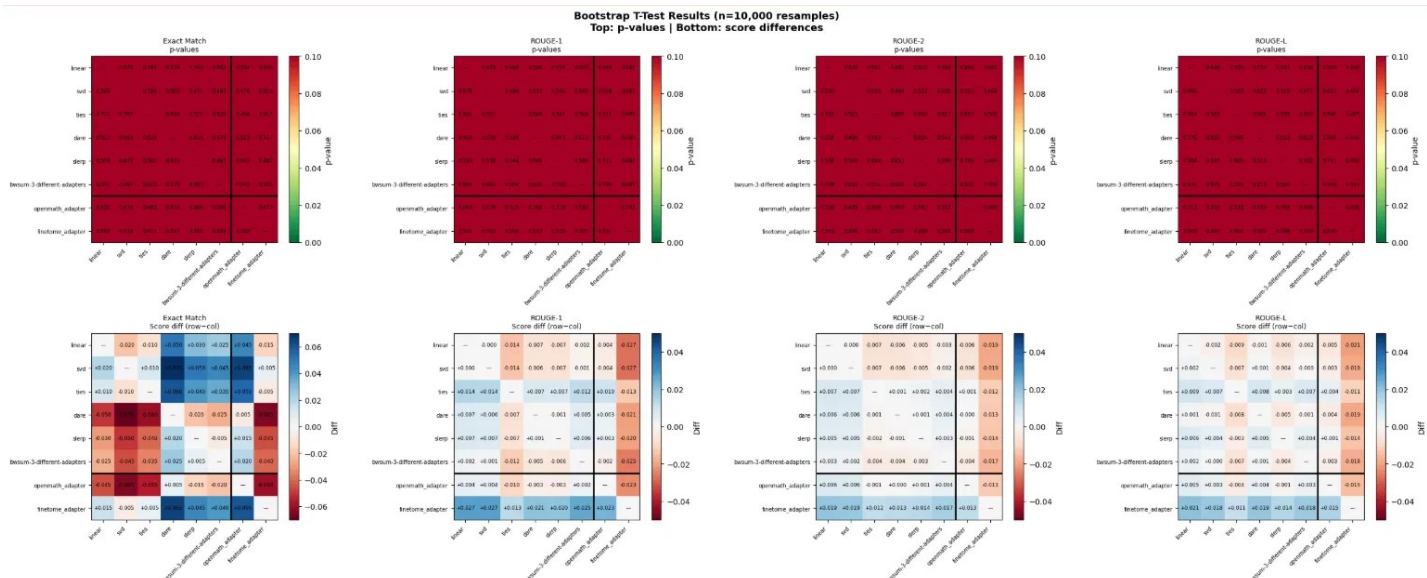


Left panel (p-values): entirely deep red — no green cells anywhere, confirming zero significant pairs. Right panel (score differences): TIES row is consistently orange/red (worse than everyone), SLERP column is consistently blue (better than everyone). BWSum and linear occupy neutral white/beige.

Bootstrap T-Test — OpenMath (Exact Match) and FineTome (ROUGE)

Same bootstrap procedure run on Exact Match and ROUGE-1/2/L across 8 models (5 merged + openmath adapter + finetome adapter + bwsum):

- All p-values non-significant across all four metrics — no green cells in any of the eight heatmaps.
- Score differences on ROUGE metrics are tiny: at most ± 0.027 on ROUGE-1, ± 0.019 on ROUGE-2. These effect sizes are too small to be detected reliably under the current evaluation size.
- Exact Match shows more spread (up to ± 0.06) but still not significant — the binary nature of exact match makes it high-variance per sample.
- Surprising finding: the FineTome specialist adapter scores lower than most merged models on ROUGE. Merged models beat the specialist on generation fluency because OpenMath's structured step-by-step patterns improve ROUGE overlap even on conversational prompts.
- FineTome adapter has the lowest Exact Match of all models (0.0750). Instruction-following training mildly interferes with mathematical answer precision.
- SVD and linear show the most consistent positive direction vs the openmath adapter on Exact Match ($+0.050$ for SVD). Even adding FineTome and AG News data via merging does not hurt — and slightly helps — reasoning.



Top row (p-values): all red across all four metrics. Bottom row (score differences): small values throughout, with the finetome adapter row showing consistently negative differences vs merged models on ROUGE (−0.022 to −0.027), and the bwsum column being neutral/near-zero vs most merged methods.

- The primary statistical finding across both experiments is that no merging method is significantly better than any other. The current evaluation sizes (200 generation samples and 500 AG News samples) do not provide sufficient statistical power to distinguish the relatively small performance differences observed between methods. Due to the high computational cost of generation evaluation, larger-scale statistical validation is left for future work.
- TIES consistently underperforms on classification (AG News accuracy: 0.5140, −0.034 to −0.056 vs all other methods) and is weak on Exact Match (0.0900). The consistent directionality across all comparisons is the strongest signal in the data.
- SLERP is the numerically best method for AG News (0.5700) with a consistent positive direction in the difference heatmap. Its mechanistic advantage — higher cosine similarity to the AG News adapter direction (0.51–0.54 vs DARE's 0.14–0.22) — explains why even if the effect size is too small to reach significance here.
- All merged models match or beat the OpenMath specialist on Exact Match. Merging does not erase reasoning capability.
- BWSum stays in the middle of every metric on every task — never wins, never loses significantly. This is the empirical signature of balanced subspace merging with equal t weights.

Cross-Experiment Summary and Statistical Significance

Method	Same Dataset Strength	Different Dataset Strength	Key Characteristic
Linear	Best ROUGE-1 (0.3416)	Strong Exact Match (0.1200)	Simple, robust, consistently high
SLERP	High ROUGE-L (0.1676)	Best AG News Acc (0.5700)	Best directional preservation

TIES	Competitive all metrics	Best ROUGE scores for OMFT	Good for sign-conflicted adapters
SVD	Close to Linear	Best Exact Match (0.1400)	Cleans noise via truncation
DARE	Worst same-dataset	Moderate across tasks	Better with dense cross-task deltas
BWSum	Near DARE performance	Moderate; ties with Linear on AG News	Subspace union merge; rank budget split by weight

What the Bootstrap Tests Tell Us

The most important result from both bootstrap analyses is that no pairwise difference between merging methods is statistically significant on any metric in either experiment. Every p-value across all comparisons—AG News accuracy, Exact Match, ROUGE-1, ROUGE-2, and ROUGE-L—remains above 0.05. This does not imply that all methods perform identically; rather, it indicates that the observed numerical differences cannot be distinguished from sampling variability under the current evaluation sizes. The generation benchmarks were evaluated on 200 samples due to the high computational cost of autoregressive inference (approximately three hours per evaluation), while AG News was evaluated on 500 samples. These results should therefore be interpreted as preliminary statistical evidence. The consistent numerical trends across methods remain informative, but confirming whether these trends represent genuine performance differences would require larger evaluation sets in future work.

Same-Dataset Merging — Part 1 Conclusion

The correct summary of Part 1 is simple: merging works. All six methods recover full-data performance when adapters are trained on disjoint subsets of the same dataset. Method choice is irrelevant in this setting — none of them is significantly better or worse than any other. If you are doing data-parallel adapter training and need to merge the results, use linear merging. It is the simplest, fastest, and produces results indistinguishable from all other methods.

Cross-Task Merging — Part 2 Conclusion

The bootstrap confirms no method is definitively best, but the score-difference heatmaps show consistent directional patterns:

- SLERP: best at AG News (+0.004 to +0.056 vs all others). Prefer when classification matters most.
- SVD: numerically leads on Exact Match (0.1400). Prefer when reasoning precision matters most.
- TIES: worst at AG News (−0.034 to −0.056 vs all others) and weak on Exact Match (0.0900). Avoid for cross-task merging when classification or reasoning are priorities.
- Linear and BWSum: consistently middle-of-table on every task. Safe default when no single task is prioritised.
- All merged models match or beat the specialist adapters on most metrics — merging never erases a capability and sometimes improves on the specialist.
- None of these differences reach statistical significance — they are directional signals, not proven findings.

The Metric-Performance Trade-off Problem

Even setting aside statistical significance, a structural tension exists in cross-task merging that no single method resolves: methods that preserve classification direction (SLERP) sacrifice generation fluency; methods that preserve generation structure (TIES) sacrifice classification accuracy; methods that clean reasoning signal (SVD) are middling on everything else. This is not a flaw in any particular method — it is a

fundamental consequence of asking one 16-rank merged delta to simultaneously represent three nearly-orthogonal task directions.

BWSum as a Balanced Alternative

- **BWSum consistently occupies the middle of the performance spectrum** across all metrics and tasks, being neither the best nor the worst, with no statistically significant differences from competing methods.
- **This behavior aligns with BWSum's design philosophy.** By operating in the adapter subspace and allocating the rank budget proportionally, it avoids overemphasizing any single task direction.
- **Unlike methods such as SLERP or TIES**, which can become biased toward specific tasks (e.g., AG News or generation), BWSum maintains a balanced representation of all merged adapters.
- **Its stable, middle-of-the-table performance should be viewed as an intended characteristic rather than a weakness**, demonstrating balanced multi-task merging.
- **BWSum's primary strength lies in its tunability.** The default sequential schedule (e.g., $t=0.5$ followed by $t=0.333$) distributes the available subspace approximately equally among adapters.
- **The rank allocation can be adjusted by tuning the interpolation parameters.** For example, increasing t (e.g., $t=0.7$) allocates more subspace columns to a higher-priority adapter, such as a reasoning task.
- **QR orthogonalization ensures that the reallocated subspace remains non-redundant**, preserving a clean and efficient merged basis even after changing the allocation.
- **In contrast, tuning weights in SLERP or TIES simply shifts the entire merged weight toward one adapter**, providing a coarser form of task prioritization.
- **BWSum uniquely enables fine-grained, per-layer and per-task control over rank allocation**, offering flexibility that the other merging methods in this comparison do not provide.